Research on the distribution and control mechanism of evapotranspiration in the Nanxiaohegou watershed based on an improved S–W model（标题）

基于改进 S-W 模型的南小河沟流域蒸散发分配及控制机制

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Abstract（摘要：使用过去时） Accurate quantification of evapotranspiration (*ET*) and its components, and identifying their controlling factors, were beneficial for the rational evaluation, planning, and management of regional water resources.（从事研究重要性）Based on continuous long-term observation data and field experiments conducted from 2016 to 2020 in the Nanxiaohegou watershed, a typical small watershed for water and soil conservation on the Loess Plateau, an improved Shuttleworth-Wallace (S-W) model was used to simulated the dynamic changes of *ET* and its components in typical plantation land. Additionally, we analyzed the coupling relationships between plant transpiration (*T*), soil evaporation (*E*), and control factors using Structural Equation Modeling (SEM).（使用的方法S-W模型和SEM方程） The results showed that: (1) The improved S–W model had good applicability for evaluating *ET* and its components in Nanxiaohe Watershed. The threshold value of soil surface resistance (r_s^s) was 50–2500 $s \cdot m^{-1}$, exhibiting an exponential relationship with the empirical function of soil surface water content (θ); moreover, the larger the sand content in soil particles, the greater the slope of the line. (2) *ET* ranged from 276.76 mm to 402.86 mm in typical plantation land, with multi-year average contributions of *T* and *E* accounting for 51.6% and 48.4% of *ET*, respectively. The patterns of *ET*, *T*, and *E* showed no significant monthly changes but drastic fluctuations during the day. The fluctuation trends of T and E largely reflected annual precipitation patterns but lagged behind rainfall. (3) SEM analysis revealed that net radiation (R_n), air temperature (T_a), and θ had the most significant impacts on *ET*, with R_n having the largest impact on *T* (total impact of 0.614) and T_a having the greatest impact on *E* (total impact of 0.426). *T* was positively correlated with *E*, with a contribution coefficient reaching 0.503.（如上3点为获得的基本结论和研究结果） Evaluating *ET* and its components based on an improved S–W model established a foundation for deeply revealing ecological and hydrological processes in arid and semiarid regions.（结论的意义——研究生态水文过程）

（该摘要属于报道性摘要，全面、简要地概括论文的目的、方法、主要数据和结论）

精准量化蒸散发（*ET*）及组分并探明其控制因子，有利于合理评价及规划管理区域水资源。根据黄土高原水土保持治理典型小流域南小河沟流域 2016—2020 年连续性长期观测数据及野外试验，基于改进的 S-W（Shuttleworth- Wallace）模型，模拟典型人工林地 *ET* 及组分动态变化，并利用结构方程模型（Structural Equation Modeling，SEM ）分析植物蒸腾（*T*）、土壤蒸发（*E*）与控制因子间的耦合关系。结果表明:（1）改进的 S-W 模型在南小河沟流域适用性较好，其中土壤表面抗阻力（r_s^s）的阈值为 50~2500 s • m^{-1}，与土壤表层含水量(θ)经验函数类型为指数型， 土壤颗粒中沙粒含量越大，线型斜率越大。（2）典型人工林地 *ET* 范围为 276.76 ~402.86 mm，多年平均 *T*、*E* 占 *ET* 的 51.6%和48.4% ，*ET*、*T* 和 *E* 规律为各月间变化不明显，但日间波动剧烈，*T* 和 *E* 波动趋势基本一致，与年降雨量变 化一致，和次降雨表现滞后性。（3）SEM 分析净辐射（R_n）、气温（T_a）、θ对 *ET* 影响最为显著，其中 R_n 对 *T* 影响最大（总影响为 0.614 ），T_a 对 *E* 影响最大（总影响为 0.426 ）；*T* 对 *E* 体现为正相关，贡献系数达 0.503 。基于改 进的 S-W 模型对 *ET* 进行评价及组分分离，为深刻揭示干旱半干旱区生态水文过程提供依据。

Key words evaporative components; improve the S-W model; SEM model; evaporation resistance; Loess Plateau; ecological hydrological processes（6个关键词，选自title和abstract）

蒸散发组分；改进 S-W 模型；SEM 模型；蒸发抗阻力；黄土高原；生态水文过程

1.Introduction（引言 使用现在时）

Evapotranspiration (*ET*) is an indispensable part of the hydrological cycle[1], playing a significant role in the water and energy balance of ecosystems. Particularly in the Loess Plateau region, *ET* accounts for more than 80% of the water loss in the system[2]. As a key factor in plant water requirements and water resource evaluation, *ET* is also an important indicator for drought monitoring. Its two components, plant transpiration (*T*) and soil evaporation (*E*), involve different biological and physical processes and have distinct dominant factors[3]. Therefore, accurately simulating *ET* and its components and studying the control mechanisms of evapotranspiration distribution are significant for the efficient planning, management, and application of water resources in the loess gully region.（研究的背景和研究的重要性）

蒸散发（*ET*）是水文循环中不可或缺的一环[1]，是生态系统中水分及能量平衡的重要组成部分，尤其在黄土高原地区，蒸散发占系统水量损失的 80%以上[2] 。*ET* 同时作为植物需水及水资源评价的关键，是干旱监测的重要指标，它的两个组成部分植物蒸腾（*T*）和土壤蒸发（*E*）有不同的生物和物 理过程以及主导因素[3] 。因此，精确模拟 *ET* 及组分并研究蒸散发分配的控制机制，从组成部分揭示 蒸散发过程，对黄土沟壑区水资源的高效规划、管理和应用等均具有重大意义。

Currently, *ET* assessment methods mainly include stable isotopes, sap flow, and eddy covariance techniques[4], which have the advantage of direct measurement without being constrained by hypothetical conditions[5]. However, they are limited in practical application becuse of high economic costs, low observation frequencies, difficulties in instrument maintenance, and narrow applicability. Hence, mathematical modeling is the primary method for *ET* assessment, with the Penman-Monteith model (P-M model) [6-7] and the Shuttleworth-Wallace model (S-W model) [8-9] being widely used for their simplicity, adaptability, and cost-effectiveness, especially in areas with complex geomorphological conditions. However, the P-M model cannot separate the components of evapotranspiration, whereas the The S-W model is better due to its ability to capture physical processes, fewer parameters with clear physical meanings, and higher simulation accuracy[10]. The improved S-W model can address the issue of calculating soil surface resistance for different soil textures[11-12] and has been proven effective in *ET* separation applications in arid and semi-arid areas.

目前 *ET* 评估方法主要包括稳定同位素、液流和涡动相关技术等方法[4]，其不受假设条件约束，具有直接测量的优势。但因经济成本较高、观测频率低、仪器难以维护、适用范围窄等因素在实际应 用中受到不同程度的限制[5] 。所以数学模型法是目前评估*ET* 的主要方法，其中 Penman-Monteith 模型（P-M 模型）[6-7]和 Shuttleworth-Wallace 模型（S-W 模型）[8-9]因结构简单、适应性强、经济方便得到广泛应用，尤其在地形地貌条件复杂区域。但由于 P-M 模型不能进行蒸散发组分分离，而 S-W 模型 因能捕获物理过程，具有参数相对较少且物理含义清晰、模拟精度较高的优点[10]，成了更好地选择。改进的 S-W 模型可解决计算不同土壤质地土壤表面抗阻力[11-12]问题，且已被证明在干旱半干旱地区有很好的 *ET* 分离应用[13]。

Regarding the control mechanisms of *ET* distribution and the interrelationships among its factors, scholars worldwide often analyze these through comparing pairwise correlation coefficients or using traditional statistical methods like multiple regression[12,14-15]. These methods only analyze the relationship between one or more independent variables and one dependent variable, failing to assess multiple independent variables' impact on multiple dependent variables. They also often overlook the autocorrelation between independent and dependent variables, leading to inconsistent or even opposite results[16-17].（Research Situation 科研现状） In recent years, Structural Equation Modeling (SEM) has emerged as an effective method to address above issues. SEM not only fully considers the correlation between independent variables but also distinguishes the impact of independent variables on dependent variables, which has been widely applied in economics and ecology[18], although its application in analyzing the control factors of evapotranspiration is still limited.（What are the key problems now目前的关键问题）

针对 *ET* 分配控制机制及其因子间的相互关系，国内外学者多通过比较两两相关的决定系数，或利用多元回归等传统统计方法进行分析[12, 14-15]，这些方法仅分析了一个或多个自变量与一个因变量之 间的关系，不能进行多个自变量对多个因变量的分析，并且往往忽略了自变量和因变量之间的自相关，导致部分结果不一致甚至相反[16-17]。近年来，结构方程模型（ Structural Equation Modeling ， SEM）成为有效解决上述问题的方法，它不仅能充分考虑自变量之间的相关关系，还能

区分自变量对 因变量的影响，被广泛应用于经济学和生态学中[18]，但目前在蒸散发控制因子分析中研究较少。

As a typical prototype observation watershed in the loess gully area, this study uses long-term continuous observation data and field experiments from the Nanxiaohegou Watershed to quantify and dynamically evaluate *ET* and its components through the improved S-W model. The SEM model is used to analyze the control mechanisms, clarifying the response process between *ET* and its components and control factors.(How we tackled the problems如何解决问题） This provides a reference for estimating plant water requirements, improving water use potential, and conducting drought monitoring and assessment research in the loess gully area of the Loess Plateau.（Assess the effectiveness of the research 评估研究成果）

南小河沟流域作为黄土沟壑区典型原型观测小流域，本文基于流域内典型人工林地长期的连续性 观测数据及野外试验，通过改进 S-W 模型对 *ET* 及组分进行量化和动态评价，利用 SEM 模型分析其 控制机制，使 *ET* 及组分与控制因子间的响应过程更加清晰，为合理的估算黄土高原沟壑区植物需水 量，提高水分利用潜力，以及开展干旱监测和评估相关研究提供参考依据。

2 Materials and Methods（材料和方法——使用过去时）

2.1 Overview of the study area（研究区概况）

The study area was located in the Nanxiaohegou Watershed, within the Xifeng District of Qingyang City, Gansu Province (35°40′43"~35°44′58"N, 107°30′8"~107°37′52"E), covering an area of 38.7 km^2. This watershed, established in 1951 by the Xifeng Station of the Loess Plateau Soil and Water Conservation Experiment, served as a typical prototype observation site for the gully region of the Loess Plateau (Figure 1). It had the distinctive topographical features of the Loess Plateau gully region, including tablelands, ridges, and valleys. The geological structure of the watershed was simple, with sandy loam soil that was prone to erosion and water loss. In recent years, with the implementation of large-scale soil and water conservation efforts in the Loess Plateau region, the vegetation coverage in the watershed had significantly increased. The main vegetation types included black locust (*Robinia pseudoacacia*), Oriental arborvitae (*Platycladus orientalis*), Chinese pine (*Pinus tabulaeformis*), Siberian apricot (*Armeniaca sibirica*), and alfalfa (*Medicago sativa*).

研究区位于甘肃省庆阳市西峰区境内的南小河沟流域（35°40′43"~35°44′58"N ，107°30′8"~107°37′52"E），流域面积为 38.7 km2 ，是黄委会西峰水土保持试验站于 1951 年建立的黄土沟壑区典型原型观测小流域（图 1）。具有典型的黄土高原沟壑区地貌特征，主要包括塬面、梁峁 坡、沟谷三大类，流域地质构造单一，土壤类型为砂壤土，抗蚀性差，易发生水土流失。近年来，随 着黄土高原地区大规模水土保持工作的开展， 流域内植被覆盖度明显增加 ，植被类型主要以刺槐 （Robinia pseudoacacia ）、侧 柏 （ Platycladus orientalis ） 、 油松（ Pinus tabulaeformis ）、山杏 （Armeniaca sibirica） 以及紫花苜蓿（Medicago sativa）等。

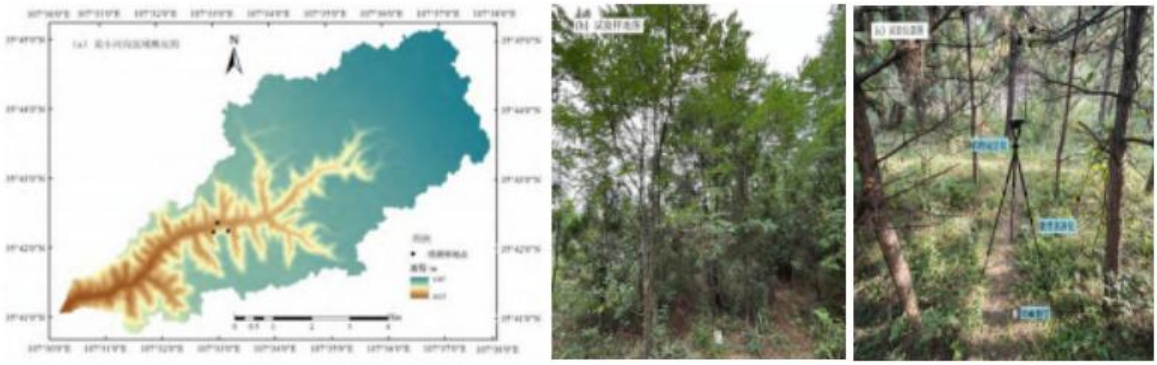


Fig. 1　Schematic diagram of study area and plot and partial test map

2.2 Data sources and processing（数据来源与处理）

Three sample plots with consistent tree ages and good growth were selected for the mixed plantations of black locust, Oriental arborvitae, and Chinese pine. Field experiments and observations were conducted within these plots from May to October in the years 2016, 2017, 2018, 2019, and 2020. The observations primarily included meteorological elements, soil moisture content, soil evaporation, and vegetation leaf area index. The plant parameters were represented by the average values of the three tree species, and the specific conditions of the tree species are presented in Table 1.

样地分别选用 3 个树龄基本一致、长势较为良好的刺槐、侧柏、油松人工乔木混交林样地。在样 地内进行野外试验布设及观测，观测时间为 2016 年、2017 年、2018 年、2019 年和 2020 年的 5—10 月，观测内容主要包含气象要素、土壤含水量、土壤蒸发、植被叶面积指数等。其植物参数以 3 种树 种平均值代替，树种具体情况见表 1。

Tab. 1　Basic information of the test site

树种	树高/cm	胸径/cm	冠幅/cm	土质	地形	坡位	坡度/（°）
刺槐	564.11±29.60	15.91±2.42	293.11±15.20	砂壤土	阶地	中	26
侧柏	499.43±27.91	17.44±3.73	211.42±11.63	砂壤土	阶地	中	22
油松	446.21±24.23	15.41±1.90	191.60±12.31	砂壤土	阶地	中	21

2.2.1 Meteorological data（气象数据）

Meteorological data were measured by three small field weather stations (Watch dog series 2000, China). The instruments, powered by solar energy, recorded indicators such as wind speed, wind direction, air temperature, humidity, total radiation, and precipitation. The weather stations were installed in open areas near the three sample plots, with an observation interval set at 1 hour. The observation results were converted into daily-scale data for subsequent analysis and calculation. To avoid missing or distorted data, the weather stations were checked and maintained approximately every week. Any missing data were rare and were replaced by data from the nearest weather station.

气象数据由 3 台小型野外气象站（Watch dog series 2000, China）进行测定。仪器设备采用太阳能 供电，观测指标包括风速、风向、气温、湿度、全辐射、降雨量等要素。气象站点的安装在 3 个样地 附近空旷处，气象站的观测时间间隔设置为 1 h ，观测结果转化为日尺度数据，以方便进行后续的分 析与计算。为避免出现数据的缺失和失真，平均每间隔一周对气象站进行检查和维护，极少部分的缺 失数据由相距最近的气象站数据代替。

2.2.2 Soil moisture content（土壤含水量）

Soil moisture content was measured using TDR (Trime TDR, Germany) probes. Experiments were conducted on the three sample plots, with two measurement points at each plot, resulting in a total of six TDR probes. The probes were buried at a location equivalent to 0.7 times the average radius of the canopy. During the installation of the probes, efforts were made to minimize disturbance to the original soil of the plantations. The TDR probes measured soil moisture at depths of 0 to 2 meters, with measurement intervals of 10 cm from 0 to 1 meter and 20 cm from 1 to 2 meters. The observation interval ranged from 1 to 3 days, and the average values were calculated. To reduce errors, observations were conducted between 8:00 and 11:00 AM each day, with three measurements at each depth in each probe. The soil moisture content of each soil layer was determined by taking the average of the three observations. Additional measurements of soil moisture content were conducted after each rainfall event.

土壤含水量采用 TDR（Trime TDR，Germany）测管进行测定。在 3 个样地上开展试验，每个样 地 2 个测点，共设置 6 个 TDR 测管，测管埋设位置为 0.7 倍冠层平均半径处。布设测管时，尽量减 少对人工林原状土壤的扰动。TDR 测管观测的土壤水分测量深度为 0~2 m，0~ 1 m 测量间隔深度为 10 cm，1~2 m 范围内间隔为 20 cm 。观测间隔时间为 1~3 d，计算取平均值。为减小误差，每次观测保 持在当天 8:00—11:00 完成，每个测管内的每个测量深度观测 3 次，土层土壤含水量取 3 次观测的均 值。每次降雨后，对土壤含水量进行加测。

2.2.3 Soil evaporation（土壤含水量）

Soil evaporation was measured using micro-lysimeters. The observations were conducted on the three sample plots, with three measurement points at each plot, resulting in a total of nine PVC tubes with a length of 25 cm and a diameter of 16 cm. During installation, the PVC tubes were first driven vertically into the soil using a soil tamper, and then the original soil was carefully removed, sealed with gauze at the bottom, and placed back in position, ensuring that the soil surface inside the micro-lysimeter was level with the tube opening. Vegetation on the soil inside the PVC tubes was removed. A wire handle was attached to the top for easy subsequent observations. Observations were conducted at 18:00 every day, with an interval of 1 to 3 days. Actual evapotranspiration was calculated using the weighing method with an electronic scale of 0.01 g precision. The difference between two weighings was considered as the soil evaporation rate during a certain period. To ensure the measurement accuracy of the micro-lysimeters, the soil should be changed every 20 days, and the mass of the PVC tube and the soil inside should be consistent with the mass of the previous observation when changing the soil.

土壤蒸发使用微型蒸渗仪（Micro-lysimeters）进行观测。在 3 个样地上进行，每个样地 3 个测 点，共设置了 9 个长度为 25 cm 直径为 16 cm 的 PVC 管。在安装时，首先用土夯锤将 PVC 管垂直砸 入土中，然后小心取出原装土，用纱布封底后放回原处，设置时确保微型蒸渗仪内的土壤表面与管口 保持一致。PVC 管内土壤上的植被需进行拔出处理。顶部装有铁丝提手， 以方便后续观测，每次观测 在当天下午 18:00 进行，观测时间间隔为 1~3 d 。实际蒸散发由称重法进行计算，称重所使用精度为 0.01 g 的电子天平，两次称量的差值即可视为一定时间内土壤的蒸发量。为了保证微型蒸渗仪的测量 精度，每隔 20 d 应换土一次，换土时（PVC 管和管内土壤）的质量保持与上次观测质量一致。

2.2.4 Leaf area index（叶面积指数）

Leaf Area Index (LAI) was measured using a plant canopy observation system (Winscanopy 2006, Canada). The measurements were conducted on the three sample plots, with two fixed observation points at each plot. After leveling the tripod, a "fisheye" camera was used to photograph three to five representative trees. The photos were processed in the Winscanopy software to obtain the LAI. The observation interval was 4 to 7 days, and the average value was calculated.

叶面积指数（LAI）使用植物冠层观测系统（Winscanopy 2006，Canada）进行观测。在 3 个样地 上进行，每块样地 2 个固定观测点，支架调平后使用“鱼眼”相机拍摄具有代表性的 3~5 棵乔木，照 片在 Winscanopy 软件中处理得到叶面积指数，观测间隔为 4~7 d，计算取平均值。

2.3Research Methods（研究方法）

2.3.1 S-W model

The Shuttleworth-Wallace model (S-W model) assumed a uniform plant canopy coverage and introduced canopy resistance and soil resistance parameters. It considered evapotranspiration as originating from two "sources": soil evaporation and plant transpiration. The total water vapor source could be represented by the latent heat flux λ*E*, and its calculation formula was as follows [10]:

Shuttleworth-Wallace 模型（S-W 模型）假设植物冠层为均匀覆盖，引入冠层阻力和土壤阻力参数，认为蒸散发来自土壤蒸发和植物蒸腾这两个“源”，总的水汽源可用潜热通量 λE 表示，其计算公式为[10]：

$$\lambda ET = \lambda E + \lambda T = C_s PM_s + C_c PM_c \quad (1)$$

in the formula,

$$E = PM_s = \frac{\Delta R + \dfrac{\rho C_p D - \Delta r_a^c \left(R - R_s\right)}{r_a^a + r_a^s}}{\Delta + \gamma\left(1 + \dfrac{r_s^s}{r_a^a + r_a^s}\right)} \quad (2)$$

$$T = PM_c = \frac{\Delta R + \dfrac{\rho C_p D - \Delta r_a^c R_s}{r_a^a + r_a^c}}{\Delta + \gamma\left(1 + \dfrac{r_s^c}{r_a^a + r_a^c}\right)} \quad (3)$$

In the formula: *ET* represented the simulated evapotranspiration rate (mm d^{-1}); λ was the latent heat of vaporization of water (MJ kg^{-1}); PM_s was the simulated soil evaporation rate (mm·d^{-1}); PM_c was the simulated plant transpiration rate (mm·d^{-1}); C_s and C_c were the proportionality coefficients for soil evaporation and plant transpiration, respectively; r_s^s was the soil surface resistance (s·m^{-1}); r_a^s was the aerodynamic resistance from the soil surface to the canopy height for latent heat (s·m^{-1}); r_s^c was the stomatal resistance of the plant canopy (s·m^{-1}); r_a^c was the aerodynamic resistance from the leaf surface to the average height of the canopy (s·m^{-1}); r_a^a was the aerodynamic resistance from the canopy height to the reference height for latent heat (s·m^{-1}); Δ was the slope of the saturation vapor pressure curve with temperature (kPa·℃$^{-1}$); *P* was the air density (1.29 kg·m^{-3}); C_p was the specific heat of air at constant pressure [1.03×10^{-3} J ·(kg · ℃)$^{-1}$]; *D* was the air saturation vapor pressure difference (kPa); γ was the psychrometric constant (0.67 hPa·k^{-1}).

式中：ET 为模拟蒸散发量（mm d-1）；λ为水的汽化潜热（MJ kg-1 ）；PMs 为模拟土壤蒸发量 （mm·d-1）；PMc 为模拟植物蒸腾量（mm·d-1）；Cs 、Cc 为土壤蒸发、植物蒸腾比例系数； rss 为土壤表面抗阻力（s·m-1）； ras 为潜热从土壤表面到冠层高度的空气动力抗阻力（s·m-1）； rsc 为植被冠层 气孔抗阻力（s·m-1）；rac 为叶片表面到冠层平均高度的空气动力抗阻力（s·m-1），raa 为潜热从冠层高度到参考高度的空气动力学阻抗（s·m-1）；Δ为饱和水汽压随温度变化的斜率（kPa·℃-1）；P为空 气密度（ 1.29 kg·m-3 ）；Cp为空气定压比热[1.03× 10-3J ·（kg · ℃) -1] ，D为空气饱和水汽压差（kPa）， γ为干湿表常数（0.67 hPa·k-1）。

In the formula: *R* represented the radiation absorbed by the canopy (W·m^{-2}), and R_S represented the radiation absorbed by the soil (W·m^{-2}). The calculation formula was as follows:

R为冠层吸收的辐射（W·m-2），RS 为土壤吸收的辐射（W·m-2 ），计算公式为：

$$R = R_n - G \tag{4}$$

$$R_s = R_{ns} - G \tag{5}$$

$$R_{ns} = R_n exp\left(-C\mathrm{LAI}\right) \tag{6}$$

In the formula: *C* was the extinction coefficient, with a value of 0.60; *G* was the soil heat flux (W·m^{-2}); R_n was the net radiation (W·m^{-2}), and R_{ns} was the net radiation absorbed by the soil (W·m^{-2}), which was calculated based on the LAI extinction principle [19].

式中：C为消光系数，取值为 0.60；G 为土壤热通量（W·m-2 ）；Rn为净辐射（W·m-2 ），Rns 为土壤吸收的净辐射（W·m-2），由 LAI 消光原理[19]求得。

2.3.2 Structural Equation Modeling (SEM)（结构方程模型）

Structural Equation Modeling (SEM) was a method based on the covariance matrix that simultaneously analyzed the causal relationships between multiple variables. It integrated factor

analysis and path analysis, allowing for the establishment, estimation, and testing of causal relationships in models. SEM could analyze the total, direct, and indirect effects of independent variables on dependent variables, making it suitable for studying the inter relationships between multiple variables. Typically, SEM used maximum likelihood estimation for parameter estimation, and the fit function of maximum likelihood estimation conformed to the χ^2 distribution. Therefore, in SEM, the χ^2 test was often used to evaluate model performance. The software Amos 24 was used to run SEM, and the model was adjusted and revised based on the test results to analyze the control mechanisms of plant transpiration (T) and soil evaporation (E).

结构方程模型（Structural Equation Modeling ，SEM）是基于协方差矩阵同时分析多个变量之间因 果关系的方法，其整合了因素分析、路径分析，可建立、估计和检验模型的因果关系， 同时可以分析 自变量对因变量的总影响、直接影响和间接影响[20]，适用于研究多个变量对多个变量之间的相互关 系。SEM 通常以最大似然估计进行参数估计，而最大似然估计的拟合函数符合X2 分布， 因此，在 SEM 中常以X2 检验来评价模型表现。利用软件 Amos 24 运行 SEM，根据检验结果对模型进行调整与 修正，分析植物蒸腾（T）和土壤蒸发（E）的控制机制。

2.4 Calculation of model improvement parameters（模型改进参数计算）

The S-W model included a large number of parameters, in addition to environmental parameters, the core of its formula was the calculation and validation of five resistance parameters: r_s^s, r_a^s, r_s^c, r_a^c, and r_a^a. Based on the research findings of other scholars and combined with the actual conditions of the study area, the research methods for the three parameters raa, rac, and ras were relatively unified. This study followed the original model method and still used the formulas from reference [10] for calculation. Additionally, the two parameters r_s^c and r_s^s adopted the methods recommended by other scholars, and the calculation methods were as follows.

S-W 模型包含了大量的参数，除环境参数外，其公式核心是rss 、ras 、rsc 、rac 、raa 5个阻力参数的计算和验证。根据其他学者的研究成果，并结合研究区实际情况，目前 raa、rac 、ras三个参数研究方法较为统一，本研究沿用原模型方法，依然参考文献[10]的公式计算。另外rsc、rss 两个参数采用其他学者推荐的方法，计算方法如下。

1.4.1 Vegetation canopy stomatal resistance(r_s^c)（植被冠层气孔抗阻力）

$r_s^c$was expressed as a function of solar radiation, water vapor deficit, air temperature, and soil moisture content, according to the Jarvis method [21], namely:

rsc 根据 Jarvis 法[21]，表示为太阳辐射、水汽亏缺、气温和土壤含水量的函数，即

$$r_s^c = \frac{r_{smin}}{LAI}\frac{F_1}{F_2F_3F_4} \quad (7)$$

$$F_1 = \frac{1+f}{f+\frac{r_{smin}}{r_{smax}}} \quad (8)$$

$$f = \frac{0.55R_{si}}{R_{sl}}\frac{2}{LAI} \quad (9)$$

$$F_2 = 1 - 0.0025D \quad (10)$$

$$F_3 = 1 - 0.0016(298 - T) \quad (11)$$

$$F_4 = \begin{cases} 1 & \theta r > \theta wp \\ \dfrac{\theta r - \theta wp}{\theta fc - \theta wp} & \theta fc \le \theta r \le \theta wp \\ 0 & \theta r < \theta wp \end{cases} \quad (12)$$

In the formula: r_{smin} represented the minimum stomatal resistance of the plant canopy ($s \cdot m^{-1}$), and r_{smax} represented the maximum stomatal resistance of the plant canopy ($s \cdot m^{-1}$). According to the research results of Cui Jianguo [22], they were taken as 170 $s \cdot cm^{-1}$ and 1400 $s \cdot cm^{-1}$, respectively; F_i (1, 2, 3, 4) denoted the stress functions for specific environmental variables; R_{sl} was the shortwave radiation absorbed by the plant ($W \cdot m^{-2}$); R_{si} was the solar radiation at the onset of photosynthesis ($W \cdot m^{-2}$); T_a was the air temperature (K); θ_r represented the soil moisture content at the plant root zone ($cm^3 \cdot cm^{-3}$, in this paper, the soil moisture content at a depth of 100~120 cm was used as a substitute); θ_{fc} was the field capacity ($cm^3 \cdot cm^{-3}$, usually taken as $0.75\theta_{sat}$); θ_{wp} was the plant wilting coefficient (0.05 $cm^3 \cdot cm^{-3}$);

式中：rsmin 表示植被冠层最小气孔抗 阻 力（ s • m-1 ）， rsmax表示植被冠层最大气孔抗阻力（s • m-1），根据崔建国[22]的研究结果分别取 170 s • cm-1 、1400 s • cm-1 ；Fi（ 1 、2 、3 、4）表示特定环 境变量的应力函数 ；Rsi 为植物吸收短波辐射（W • m-2 ） ；Rsl为光合作用开始时的太阳辐射 （W • m-2 ）；Ta 为空气温度（K）； θ r 表示植物根系土壤含水量（cm3 • cm-3 ，本文取深度为 100~ 120 cm 的土壤含水量代替），θfc为田间持水量（cm3 • cm-3 ，通常取 0.75 θ sat），θwp为植物凋萎系数（0.05cm3 • cm-3 ）；

2.4.2 Soil surface resistance(r_s^s)（土壤表面抗阻力）

r_s^s adjusted the movement of water vapor from the interior of the soil to the soil surface and had a strong dependence on the moisture content of the upper soil layer. Studies have shown that there was a functional relationship between r_s^s and the soil surface moisture content θ [23]. That is,

rss 调节水蒸气从土壤内部移动到土壤表层，与上层土壤水分有很强的依赖性。研究表明 rss 和土壤表层含水量 θ 间存在函数关系[23]。即，

$$r_s^s = a_1(\frac{\theta_{sat}}{\theta})^{b_1} + c_1 \quad (13)$$

In the formula: θ represented the soil surface moisture content ($cm^3 \cdot cm^{-3}$). Due to the complexity of the soil surface, the soil moisture content at a depth of 10 cm was used as a substitute in this paper. θ_{sat} was the soil saturated moisture content (0.41 $cm^3 \cdot cm^{-3}$), and a_1, b_1, c_1 were parameters that needed to be calibrated in the model.

式中：θ 表示土壤表层含水量（cm3 cm-3 ），由于土壤表面的复杂性，本文取深度为 10 cm的土壤水分含水量代替），θsat为土壤饱和含水量（0.41cm3 ·cm-3）， a1、b1、c1为模型中需要率定的参数。

2.5 Model evaluation methodology（模型评价方法）

The purpose of model evaluation was to determine whether the model met the research requirements. In this paper, the Root Mean Square Error (RMSE), Coefficient of Determination (R^2), and Nash-Sutcliffe Efficiency Coefficient (NS) were used to evaluate the accuracy of the model simulation. The calculation formulas and evaluation criteria are detailed in reference [2].

模型评价的目的是判断模型是否满足研究要求。 本文使用均方根误差（RMSE）、决定系数（R2 ）、纳什效率系数（NS）对模型模拟精度进行评价，计算公式及评判标准详见文献[2]。